

A proof that $C'(Q)$ is the inverse supply function in a two-good QL economy

This is MWG 10.C.3. (MWG 10.C.4 follows as a corollary.) Consider the problem of choosing a profile of outputs $q = (q_1, \dots, q_J) \geq 0$ to minimize aggregate cost.¹ Recall we assume here that each c_j is *strictly convex* as well as C^1 with $c'_j > 0$. Since the c_j are strictly convex, the solution is unique.

$$(\text{ACM}) \quad C(Q) = \min_{q \geq 0} \sum c_j(q_j)$$

$$\text{s.t.} \quad \sum q_j \geq Q.$$

Suppose that C is continuously differentiable. The point is to show that $C'(Q)$ is the *inverse supply function*. Recall that $s_j(p)$ is firm j 's supply and the total supply is $S(p) = \sum_j s_j(p)$. It suffices to show that $S(C'(Q)) = Q$. First set $Q > 0$. (This ensures that Slater's constraint qualification holds.)

First, the FOCs for the ACM problem—which are sufficient here, as well as necessary—are that there is a $\lambda^* \geq 0$ such that for $j = 1, \dots, J$,

$$c'_j(q_j) - \lambda^* \geq 0, \quad q_j[c'_j(q_j) - \lambda^*] = 0, \quad q_j \geq 0.$$

Second, by the Envelope theorem, $C'(Q) = \lambda^*$.

[Why? For $Q > 0$, the solution $\tilde{q}$ to the ACM must involve positive output for at least one firm. Call that firm 1. Noting that the constraint in (ACM) holds as an equality since the costs are strictly increasing, we can write the ACM as $\min c_1(Q - \sum_2^J q_j) + \sum_2^J c_j(q_j)$ where the minimization is over $(q_2, \dots, q_J) \geq 0$ —uniqueness implies that the constraint $q_1 \geq 0$ does not bind—and now apply the Milgrom-Segal Envelope Theorem to find $C'(Q) = c'_1(\tilde{q}_1)$; by the FOC and $\tilde{q}_1 > 0$ at the solution, $c'_1(\tilde{q}_1) = \lambda^*$, so $C'(Q) = \lambda^*$.]

Substitute $C'(Q)$ for λ^* in the FOCs. Now these are the *same* as the FOCs for the profit maximization problems at $p = C'(Q)$. The FOCs for these problems, to repeat, are sufficient as well as necessary. These facts imply that the point $\tilde{q} = (s_1(C'(Q)), \dots, s_J(C'(Q)))$ solves (ACM) at Q . So

$$S(C'(Q)) = \sum_j s_j(C'(Q)) = Q \tag{1}$$

as asserted. The first equality is the definition of supply at $p = C'(Q)$, the second asserts that the constraint in (ACM) holds as an equality at a solution. For $Q = 0$ take $Q \rightarrow 0$ in (1).

Remark 1: As a corollary, the aggregate supply, $S(p)$, is the solution to $\max_{Q \geq 0} pQ - C(Q)$. (Why?) There is therefore a “representative” firm—a hypothetical firm which, when endowed with the aggregate production set, generates the economy's aggregate supply. This fact does not depend on what the technologies of the individual firms are and extends to general production sets beyond two-good economies: see MWG Prop 5.E.1(ii) (#8 in ASMT V-A). The associated aggregate production function is $F(z) = C^{-1}(z)$.

Remark 2: Under strict convexity on each cost function you can show that if $q^t = (q_1^t, \dots, q_J^t)$ solves the problem at $Q = Q_t$, $t = 0, 1$, $Q_1 > Q_0$, then $q^1 \geq q^0$; indeed $q^1 > q^0$ (but not $q^1 \gg q^0$ in general). It follows from MWG 3.G.4; just change the notation: replace x_i with q_j , u_i with $-c_j$, and $p \cdot x = w$ with $\mathbf{e} \cdot q = Q$, where $\mathbf{e} = (1, \dots, 1)$.

¹No firm cares about this problem. But it might be that firms are led, as if by an invisible hand, to solve it.